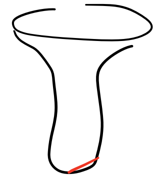
 $= \Omega$

We need to transform the PDE:

$$\frac{\partial q}{\partial t} = \frac{\partial}{\partial x}\left(\frac{\alpha_x}{q^2}\frac{\partial q}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\alpha_y}{q^2}\frac{\partial q}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\alpha_z}{q^2}\frac{\partial q}{\partial z}\right)$$

in $(x,y,z) \in \Omega$ where Ω describes the crypt geometry.

We wish to transform this PDE to (θ, z) coordinates using the following parameterization:

generate Ω

$$\vec{X}(\theta,z) = \begin{bmatrix} r(\theta,z)\cos(\theta) \\ r(\theta,z)\sin(\theta) \\ z \end{bmatrix} \qquad (\theta,z) \in [0,2\pi]\times[0,L]$$

where, for a normal stati formed crypt,

$$r(\theta,z) = r_b\left(1-e^{-az}\right) + r_t e^{a(z-L)}.$$

Here, we have

$$r_b = \frac{41}{2\pi}, \quad r_t = \frac{10}{\pi}, \quad L = 78, \quad a = 0.45$$

- In order to transform this PDE to (θ,z) coordinates, we need to compute the metric tensor g, where

$$g := \begin{bmatrix} g_{\theta\theta} & g_{\theta z} \\ g_{z\theta} & g_{zz} \end{bmatrix} = \begin{bmatrix} \frac{\partial \vec{X}}{\partial \theta} \cdot \frac{\partial \vec{X}}{\partial \theta} & \frac{\partial \vec{X}}{\partial \theta} \cdot \frac{\partial \vec{X}}{\partial z} \\ \frac{\partial \vec{X}}{\partial z} \cdot \frac{\partial \vec{X}}{\partial \theta} & \frac{\partial \vec{X}}{\partial z} \cdot \frac{\partial \vec{X}}{\partial z} \end{bmatrix}$$

With $r = r(\theta, z)$, we have

$$\frac{\partial \vec{X}}{\partial \theta} = \begin{bmatrix} \frac{\partial r}{\partial \theta}\cos(\theta) - r\sin(\theta) \\ \frac{\partial r}{\partial \theta}\sin(\theta) + r\cos(\theta) \\ 0 \end{bmatrix}, \qquad \frac{\partial \vec{X}}{\partial z} = \begin{bmatrix} \frac{\partial r}{\partial z}\cos(\theta) \\ \frac{\partial r}{\partial z}\sin(\theta) \\ 1 \end{bmatrix}$$

which gives

- $$\begin{aligned} g_{\theta\theta} &= (r_\theta\cos(\theta) - r\sin(\theta))^2 + (r_\theta\sin\theta + r\cos\theta)^2 \\ &= r_\theta^2\cos^2\theta - 2rr_\theta\sin\theta\cos\theta + r^2\sin^2\theta + \dots \\ &\qquad r_\theta^2\sin^2\theta + 2rr_\theta\sin\theta\cos\theta + r^2\cos^2\theta \\ &= r_\theta^2 + r^2 \end{aligned}$$

- $$\begin{aligned} g_{\theta z} &= (r_\theta\cos(\theta) - r\sin(\theta))(r_z\cos(\theta)) + \dots \\ &\qquad (r_\theta\sin(\theta) + r\cos(\theta))(r_z\sin(\theta)) \\ &= r_\theta r_z\cos^2\theta - rr_z\sin\theta\cos\theta + \dots \\ &\qquad r_\theta r_z\sin^2\theta + rr_z\cos\theta\sin\theta \\ &= r_\theta r_z \end{aligned}$$

- $g_{zz} = r_z^2 \cos^2\theta + r_z^2 \sin^2\theta + 1^2$

$$= r_z^2 + 1$$

Hence, the metric tensor is

$$g = \begin{bmatrix} r_\theta^2 + r^2 & r_\theta r_z \\ r_\theta r_z & r_z^2 + 1 \end{bmatrix}.$$

The determinant of this tensor is

$$|g| = (r_\theta^2 + r^2)(r_z^2 + 1) - r_\theta^2 r_z^2$$

$$= r_\theta^2 r_z^2 + r_\theta^2 + r^2 r_z^2 + r^2 - r_\theta^2 r_z^2$$

$$= r_\theta^2 + r^2(r_z^2 + 1),$$

which gives the following inverse matrix:

$$g^{-1} := \begin{bmatrix} g^{-\theta\theta} & g^{-\theta z} \\ g^{-z\theta} & g^{-zz} \end{bmatrix} = \begin{bmatrix} \frac{r_z^2+1}{|g|} & \frac{-r_\theta r_z}{|g|} \\ \frac{-r_\theta r_z}{|g|} & \frac{r_\theta^2 + r^2}{|g|} \end{bmatrix}$$

- We can transform this PDE to (θ, z) using the Laplace-Beltrami operator:

$$\Delta_{LB} f = \frac{1}{\sqrt{|g|}} \nabla_{\vec{u}} \cdot \left(\sqrt{|g|}\, g^{-1} \nabla_{\vec{u}} f \right)$$

where $\vec{u}$ is any coordinate system in 2-D. In our case, we have non-linear diffusion so

$$\Delta_{LB} q = \frac{1}{\sqrt{|g|}} \nabla_{\theta z} \cdot \left(\frac{\sqrt{|g|}}{q^2} \vec{\alpha} \cdot g^{-1} \nabla_{\theta z} q \right)$$

where $\vec{\alpha} = \langle \alpha_\theta, \alpha_z \rangle$. We can expand this equation to the following PDE

$$\frac{dq}{dt} = \Delta_{LB} q$$

$$\Rightarrow \frac{\partial q}{\partial t} = \frac{1}{\sqrt{|g|}} \nabla_{\theta z} \cdot \left(\frac{\sqrt{|g|}}{q^2} \vec{\alpha} \cdot g^{-1} \nabla_{\theta z} q \right)$$

$$\Rightarrow \frac{\partial q}{\partial t} = \frac{1}{\sqrt{|g|}} \nabla_{\theta z} \cdot \left(\frac{\sqrt{|g|}}{q^2} \begin{bmatrix} \alpha_\theta \\ \alpha_z \end{bmatrix} \cdot \frac{1}{|g|} \begin{bmatrix} r_z^2+1 & -r_\theta r_z \\ -r_\theta r_z & r_\theta^2 + r^2 \end{bmatrix} \begin{bmatrix} \partial q/\partial\theta \\ \partial q/\partial z \end{bmatrix} \right)$$

$$\Rightarrow \frac{\partial q}{\partial t} = \frac{1}{\sqrt{|g|}} \nabla_{\theta z} \cdot \left(\frac{1}{\sqrt{|g|}\, q^2} \begin{bmatrix} \alpha_\theta (r_z^2+1) \frac{\partial q}{\partial \theta} - \alpha_\theta r_\theta r_z \frac{\partial q}{\partial z} \\ -\alpha_z r_\theta r_z \frac{\partial q}{\partial \theta} + \alpha_z (r_\theta^2 + r_z^2) \frac{\partial q}{\partial z} \end{bmatrix} \right)$$

$$\Rightarrow \frac{\partial q}{\partial t} = \frac{1}{\sqrt{r_\theta^2 + r^2(r_z^2+1)}} \cdot \frac{\partial}{\partial \theta} \left(\frac{\alpha_\theta (r_z^2+1) \frac{\partial q}{\partial \theta} - \alpha_\theta r_\theta r_z \frac{\partial q}{\partial z}}{q^2 \sqrt{r_\theta^2 + r^2(r_z^2+1)}} \right) + \cdots$$

$$\frac{1}{\sqrt{r_\theta^2 + r^2(r_z^2+1)}} \cdot \frac{\partial}{\partial z} \left(\frac{-\alpha_z r_\theta r_z \frac{\partial q}{\partial \theta} + \alpha_z (r_\theta^2 + r_z^2) \frac{\partial q}{\partial z}}{q^2 \sqrt{r_\theta^2 + r^2(r_z^2+1)}} \right)$$

• we wish to numerically solve this PDE for $q(\theta, z, t)$ subject to the following boundary conditions:

$$
\begin{cases}
q(0, z, t) = q(2\pi, z, t) & \forall (z, t) \\
\nabla_{\theta z} q \Big|_{z=1} = 0 \\
q(\theta, 8z, t) = 1
\end{cases}
$$

Pseudo Code

For $t \in [t, t+\Delta t]$

① Solve $\frac{\partial q}{\partial t} = \Delta_{LB} q$ in (θ, z)

② For each z, we know $q(\theta, z, t)$

in $r(\theta, z)$.

③
$$r(\theta, 1, t) \xrightarrow{\frac{\partial \phi}{\partial t} = D\phi} r(\theta, 1, t+\Delta t)$$
$$r(\theta, 2, t) \longrightarrow$$
$$\vdots$$
$$r(\theta, 82, t) \longrightarrow r(\theta, 82, t+\Delta t)$$
same

④ Update $\vec{X}(\theta, z)$ by interpolation

$$\frac{\partial q}{\partial t} = \frac{\partial}{\partial s}\left(\frac{\alpha}{q^2}\frac{\partial q}{\partial s}\right) \qquad s \in [1, 82]$$

small

$$\frac{\partial q}{\partial t} = \frac{\partial}{\partial x}\left(\frac{\alpha_x}{q^2}\frac{\partial q}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\alpha_y}{q^2}\frac{\partial q}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\alpha_z}{q^2}\frac{\partial q}{\partial z}\right)$$

$$\frac{\partial q}{\partial t} = \nabla_{xyz} \cdot \left(\frac{1}{q^2}\vec{\alpha} \cdot \nabla_{xyz} q\right) \qquad z \in [1, 82]$$

$r_0 = \text{crypt}$

<u>Crypt</u>: $\vec{X}(\theta, z) = \begin{bmatrix} r(\theta,z)\cos(\theta) \\ r(\theta,z)\sin(\theta) \\ z \end{bmatrix}$

$\vec{\alpha} = (\alpha_\theta, \alpha_z)$

$$\frac{\partial q}{\partial t} = \Delta_{LB}\, q$$

Now we want the parametrization to be

$$\vec{X}(\theta,z;c) = \begin{bmatrix} f(\theta,z;c)\, r(\theta,z)\cos(\theta) \\ f(\theta,z;c)\, r(\theta,z)\sin(\theta) \\ z \end{bmatrix},$$

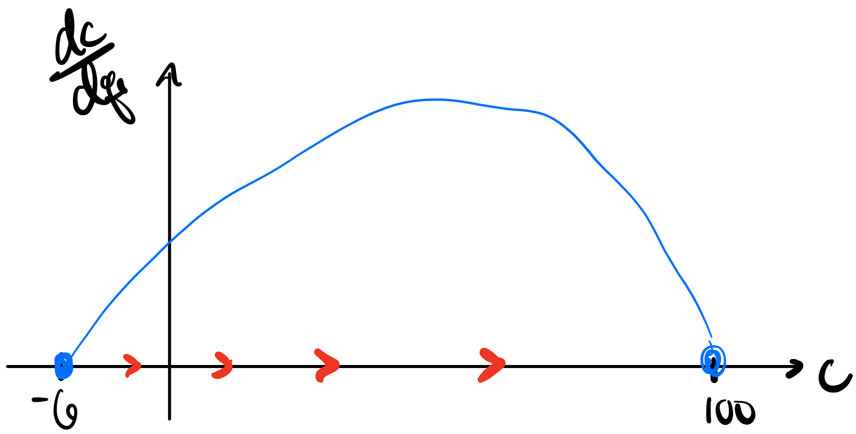
where

$$r(\theta,z) = r_b\left(1-e^{-\alpha z}\right) + r_t e^{-\alpha(z-L)}$$

and

$$f(\theta,z;c) = 1 - \frac{1-2\sin^2(\theta)}{1+e^{-\beta(z-c)}} \qquad \begin{cases} z \gg c : f \to 2\sin^2(\theta) \\ z \ll c : f \to 1 \end{cases}$$

So... $c \in [-5, 100]$ depends on $q_r = \dfrac{\iint_\Omega q(\theta,z)\,dS}{\iint_\Omega q_0(\theta,z)\,dS}$



$$\frac{dc}{dq_r} = k(c+6)(c-100) \longrightarrow \boxed{c(q_r) = \frac{100+6\left(\frac{c_0-100}{c_0+6}\right)e^{106kq_r}}{1-\left(\frac{c_0-100}{c_0+6}\right)e^{106kq_r}}}$$